RV-Sparse Coding Challenge Submission

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Repository Link

🔗 https://github.com/kirito632/rv-sparse-challenge

Implementation Highlights

- **Zero Dynamic Memory Allocation:** Strictly compliant with the challenge constraint. Executed entirely within caller-provided buffers.
- **One-Pass CSR Extraction:** Implemented a cache-friendly approach that extracts Compressed Sparse Row (CSR) structure in a single traversal, halving the memory traffic compared to multi-pass counting.
- **Auto-Vectorization Friendly SpMV:** Hoisted row bounds (`row_start` / `row_end`) into local variables to eliminate pointer aliasing, empowering compiler auto-vectorization for the inner reduction loop.
- **Validated Correctness:** Passed 100 randomized dense-to-sparse verification tests using mixed absolute/relative floating-point tolerance logic in ISO C.

Build & Run

```
gcc -o run challenge.c -lm
./run
```